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NeuroAgent

A benchmark and an agent stack for
evidence-driven neurological diagnosis

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Outline

1. Motivation

2. Benchmark

3. Validation

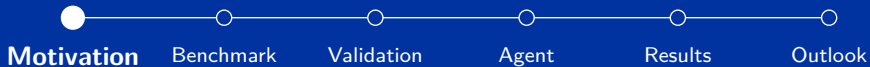
4. Agent

5. Results

6. Outlook

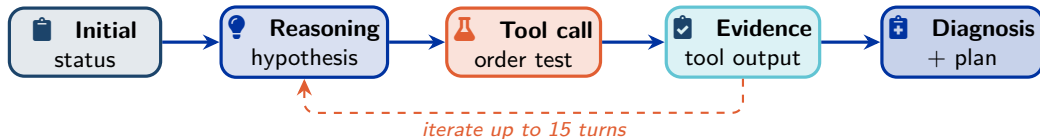


Motivation



Project goal

Build a benchmark *and* an agent pipeline that measure (and improve) **evidence-driven** clinical reasoning. Neurology is an ideal test bed (MRI, EEG, EMG, CSF, labs).



Realistic cases

500+ simulated neurology cases that mirror real practice: the agent uncovers diagnostic evidence step by step through tool calls, the way a clinician orders tests.

Multi-paradigm agents

A ladder of variants: vanilla LLM → tool use → SFT on golden traces → GRPO → graph-structured reasoning.

Expert validation

Every case reviewed by ≥ 3 clinicians on a custom platform. **Inter-rater agreement** = gold standard.

Medical LLM benchmarks reward trivia, not reasoning

How LLMs are evaluated today

- Datasets: MedQA Jin et al. (2021), MedMCQA Pal et al. (2022), USMLE-style sets
- Static multiple-choice questions
- Every relevant fact is in the stem
- Exactly one correct option
- Rewards recall + pattern matching

How clinicians actually work

- Start with a messy chief complaint
- *Decide* which tests to order
- Weigh cost, risk, time, stewardship
- Resist red herrings and anchoring
- Iterate evidence → hypothesis

The exam pre-solves the hardest step of diagnosis:
deciding *what evidence is worth looking at.*

The cost of measuring the wrong thing

Frontier LLMs reportedly match or beat clinicians on medical exams [Singhal et al. \(2023\)](#), yet real deployments tell a more complicated story.

What multiple-choice scoring **cannot** measure:

- **Tool selection:** did the agent order the right test at the right time?
- **Tool parameterisation:** with the right panel, sequence, or region?
- **Cost and stewardship:** did the agent order three MRIs when one would suffice?
- **Red-herring resistance:** did it chase the obvious finding past the contradicting one?
- **Reasoning trace:** can a clinician follow the logic from chief complaint to plan?

MedCaseReasoning [Wu et al. \(2025\)](#) makes the same point on reasoning fidelity; we extend it with *tool use*.

Two complementary contributions

NeuroBench

- **365** synthetic from per-condition YAML specs
- **151** seeded from real PMC reports (CC-BY 4.0) [Wu et al. \(2025\)](#); [Zhao et al. \(2022\)](#)
- Evidence moved out of the HPI (history of present illness) into **tool outputs**
- Rich ground truth: differential, optimal sequence, red herrings, contraindicated actions

NeuroAgent

ReAct [Yao et al. \(2023\)](#) tool-augmented agent under five progressively richer paradigms:

- Vanilla LLM (no tools): floor
- Tool-augmented (12 diagnostic tools)
- + SFT on golden trajectories
- + GRPO [Shao et al. \(2024\)](#) (*planned*)
- + graph-based structured reasoning (*planned*)

Closest prior work: AgentClinic

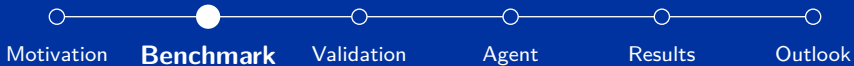
AgentClinic [Schmidgall et al. \(2024\)](#) pioneered agentic clinical evaluation; NeuroBench targets a complementary axis, **evidence-driven tool use**.

Dimension	AgentClinic (npj Dig. Med. 2026)	NeuroBench (ours)
Cases	122 (MedQA + NEJM); leakage risk	516 (synthetic + PMC-seeded); low leakage
Clinical scope	General USMLE breadth	Neurology, in depth
What it probes	Doctor–patient dialogue, history-taking	Tool selection on a stripped HPI
Tool model	Measurement agent returns values	12 typed tools, follow-up branching
Cost & stewardship	<i>None</i> (their stated future work)	Medicare-PFS cost per tool call
Scored on	Diagnosis label only	Dx + optimal / contraindicated actions
Replay	LLM agents: non-deterministic	Static cache: deterministic, GPU-free
Beyond evaluation	Benchmark only	Benchmark + SFT → GRPO training

Complementary, not competing: dialogue-driven history-taking (AgentClinic) vs. evidence-driven tool use (ours).



Benchmark



NeuroBench at a glance

Cases

516

20 conditions

Tools

12

Pydantic-typed outputs

Follow-ups

3,307

pre-cached outputs

What it is. 500+ simulated neurology patients for evaluating **tool-augmented** LLM agents on clinical reasoning, not pattern recognition.

Distinctive properties.

- **20 conditions** spanning neuromuscular, vascular, demyelinating, dementia, and more
- 12-tool schema with Medicare-PFS cost tracking
- 3 difficulty tiers; mix of synthetic and PMC-seeded

Two generation pipelines, one schema

Synthetic pipeline $n = 365$

Per-condition YAML spec defines disease, expected tool outputs, and reasoning rubric.

LLM receives:

1. Pydantic JSON schema (NeuroBenchCase)
2. Condition YAML
3. Difficulty tier + encounter type

Real-PMC-seeded pipeline $n = 151$

Seed: a published PMC case report ([Wu et al. \(2025\)](#)).

The LLM is instructed to:

1. Preserve the clinical trajectory
2. **Strip** diagnostic results out of the HPI
3. Move them into structured tool outputs
4. Add red herrings at higher difficulties

Both pipelines emit the same schema. Case IDs starting with R are PMC-seeded (e.g. ALS-RS11); others are synthetic.

Anatomy of a case: building the patient picture

What the agent sees at turn 1

Demographics	Chief complaint	HPI narrative	Vitals	Neuro. exam
Past medical hx	Medications	Allergies	Family hx	Social hx

Revealed only through tool calls: the 12-tool catalogue

 analyze_brain_mri	 analyze_eeg	 analyze_ecg	 interpret_labs
 analyze_csf	 order_ct_scan	 order_echocardiogram	 order_cardiac_monitoring
 order_advanced_imaging	 order_specialized_test	 search_medical_literature	 check_drug_interactions

Together these compose the complete patient; the agent must call the right tools to fill in the orange blocks.

Case study: ALS-M01 at a glance

Field	Value
Case ID / condition	ALS-M01: ALS (bulbar-onset)
Difficulty / encounter	moderate (concurrent cervical spondylosis), outpatient
Patient	62 y female, right-handed, retired teacher, BMI 23.1
Chief complaint	Progressive dysarthria + dysphagia + limb weakness over 8 months
Ground-truth Dx	ALS bulbar-onset, moderate stage; ICD-10 G12.21
Initial tool outputs	MRI brain + cervical spine, labs, EMG/NCS
Follow-ups ($n = 6$)	PFTs, genetic panel, CSF NfL, repeat EMG, MDC referral, riluzole check

Pedagogic point: tests whether the agent commits to neurosurgical referral for the cervical spondylosis, or recognises that bulbar findings plus supra-cervical EMG denervation cannot be explained by cervical disease.

What the agent sees at turn 1

Output of format_patient_info()

```
1 Patient: 62 yo female. CC: progressive dysarthria, dysphagia, and
2   bilateral limb weakness over 8 months.
3 HPI: 62-yo right-handed woman with 8-month progressive dysarthria,
4   dysphagia, and limb weakness. PCP ordered a cervical MRI showing
5   moderate spondylosis; cervical myelopathy raised as etiology...
6 PMH: HTN; hypothyroidism; osteoporosis; remote right CTS released 5y ago
7 Exam:
8   Motor: deltoids 4/5, biceps 4+/5, hand 4-/5 R, 3+/5 L. Visible
9   fasciculations bilateral UE proximal muscles.
10  Reflexes: hyperreflexia 3+; plantars extensor (Babinski +);
11  Hoffman sign + bilaterally.
12  Add'l: fasciculations in tongue, deltoids, left thenar...
13 Vitals: BP 128/78, HR 72, T 36.8 C, RR 14, SpO2 97%
14
```

No MRI report, no labs, no EMG, no CSF; the agent decides what to order.

Hidden inside the case: what tools return

analyze_brain_mri (initial)

Bilateral CST T2 hyperintensity in posterior limb of internal capsule. Cervical spondylosis, C5–C6 stenosis, *no cord signal change*.

Confidence: 0.0; impression reads “clinical correlation recommended.”

interpret_labs (NM panel)

CK **748 U/L** (H), Aldolase 8.2 (H, mild). TSH, B12, ESR, CRP normal. Anti-GM1 neg. SPEP no M-spike. AR-CAG 22 (*normal*).

order_specialized_test (EMG/NCS)

Active denervation + chronic reinnervation across **bulbar, cervical, thoracic**; early lumbosacral changes. Sensory NCS normal. No conduction block.
EI Escorial revised: clinically definite ALS.

analyze_csf (follow-up)

OP 14 cmH₂O; WBC 2; protein 58 (mildly elevated). **NfL 8,420 pg/mL** (ref <1,000). Oligoclonal bands absent. Cytology negative.

Each output is a Pydantic model with full panel structure and per-test abnormality flags.

Off-pathway tool calls stay in character

A case pre-caches outputs only for its diagnostic pathway, yet the agent may call any of the 12 tools. A second **fallback tier** answers the rest, so the simulation never returns a “no data available” error.

What an off-pathway tool returns

- **Non-contributory normal** (87%): a properly structured normal report for that modality
- **Incidental finding** (13%): abnormal but not the diagnosis, such as chronic small-vessel change, a benign cyst, or trace valvular regurgitation

Incidental rate calibrated to the imaging-incidentaloma literature (15–30% of scans).

2,813 fallback outputs, drawn deterministically per (case, tool) and flagged for the evaluator, so over-ordering is always scored.

Why it matters

Over-ordering is no longer a free dead-end: it costs money, burns turns, and forces the agent to **dismiss red herrings it created itself**.

Ground truth grades the *reasoning path*, not just the answer

Optimal action sequence (excerpt)

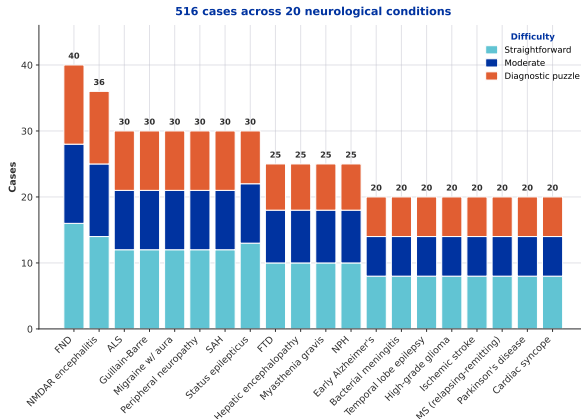
- | | |
|-----------------------------------|-------------------|
| 1. MRI brain + cervical spine | <i>required</i> |
| 2. Labs: CK, anti-GM1, SPEP, TSH | <i>required</i> |
| 3. EMG/NCS across 4 regions | <i>required</i> |
| 4. Pulmonary function (FVC, SNIP) | <i>required</i> |
| 5. ALS genetic panel | <i>acceptable</i> |
| 6. Riluzole + ALS MDC referral | <i>required</i> |

Red herrings encoded in the case

- **C5–C6 stenosis** pulls toward neurosurgery. Correct: no cord signal change; bulbar findings unexplained by cervical pathology.
- **Remote right CTS** suggests entrapment for hand weakness. Correct: EMG denervates non-median muscles bilaterally.

Each step stores `expected_finding` and `tool_parameters`, so the evaluator grades not just *whether* the right test was ordered but *how* it was parameterised.

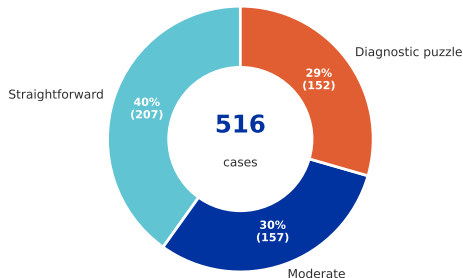
20 conditions across neurology subspecialties



Coverage spans neuromuscular (ALS, MG, GBS, PN), epilepsy/status, vascular (stroke, SAH), dementia (AD, FTD, NPH), demyelinating (MS, NMDAR), infectious/metabolic, headache, neuro-oncology, movement, and FND.

Difficulty calibration

Difficulty distribution



Straightforward (40%)

Classic textbook presentation. An intern should get this right.

Moderate (30%)

Atypical features or subtle findings. 1–2 **red herrings** (incidental labs, borderline imaging, confounding meds).

Diagnostic puzzle (30%)

Initial presentation mimics a different condition. Key diagnostic clue **buried in follow-up results**.



Validation



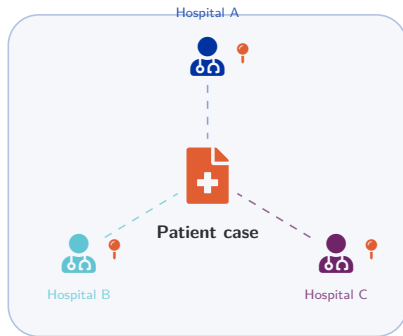
Synthetic data alone is not enough

The benchmark is only as credible as the cases inside it.

LLM-generated trajectories can be:

- **Internally inconsistent** (lab values that contradict the diagnosis)
- **Clinically implausible** (mis-staged disease, wrong-age presentation)
- **Mis-keyed red herrings** that aren't actually distracting

Our approach: every case reviewed by ≥ 3 independent clinicians on a dedicated platform, with field-level annotations and inter-rater agreement tracking.



The review platform: Overview tab

What it is

Separate FastAPI + Vite app, port 8889, gated by X-Reviewer-Code header. File-based persistence under data/review/annotations/. Reuses the same NeuroBenchCase schema.

Four tabs

- **Overview:** per-reviewer progress dashboard
- **Cases:** 516 cases, field-level annotation
- **Methodology:** public showcase
- **Admin:** aggregate agreement views

APP SCREENSHOT

Overview tab: reviewer greeting, per-condition progress bars, recent activity feed, status pie (pending / in-progress / approved / needs-changes).

Annotation workflow: field-level, gesture-driven

Status flow (per reviewer, per case)

pending → in_progress (*auto on first annotation*) → needs_changes / approved

Annotation gesture

Hover any AnnotatableField → Comment pill in margin → Radix Popover with severity (note / issue / error), textarea, and Cmd+Enter save. Annotated fields get a left border coloured by highest severity.

Filesystem isolation

One JSON file per (reviewer, case) pair, so a reviewer endpoint cannot return another reviewer's data.

APP SCREENSHOT

Cases tab: annotated MRI report with a margin Comment pill open into a Radix Popover with severity (note / issue / error), textarea, and Cmd+Enter save.

Admin tab: aggregate views over all reviewers

Four aggregate views

- **Inter-rater agreement:** per-case verdict + Cohen's κ
- **Reviewer progress:** cases over time
- **Field hotspots:** most *error*-tagged fields
- **Side-by-side diff:** compare 3 reviewers on the same case

Output: a trusted dataset

Cases with ≥ 3 *approved* verdicts and high κ graduate into the locked benchmark.

APP SCREENSHOT

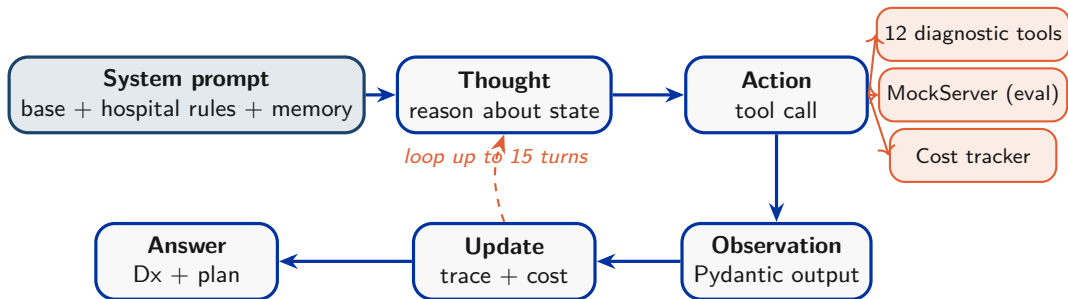
Admin dashboard: inter-rater agreement heatmap (rows = cases, cols = reviewers) with Cohen's κ card; reviewer-progress sparkline; field-hotspot bar chart.



Agent



NeuroAgent architecture: ReAct loop up to 15 turns



Built on ReAct [Yao et al. \(2023\)](#); tool-augmentation pattern from Toolformer [Schick et al. \(2023\)](#). In evaluation mode, MockServer replays pre-generated tool outputs, with no GPU needed.

The 12 diagnostic tools

Tool	Returns	Cases
interpret_labs	lab panel, reference ranges, abnormality flags	516
analyze_brain_mri	MRI report, region-tagged abnormalities	376
analyze_eeg	EEG: background, epileptiform features, focality	362
analyze_ecg	ECG: rhythm, intervals, ischemia, dysrhythmia	314
analyze_csf	CSF: cells, protein, glucose, NfL, OCB, cytology	302
search_medical_literature	curated guideline / paper snippets	216
check_drug_interactions	interaction profile + contraindications	216
order_specialized_test	EMG/NCS, biopsy, autoimmune panels	169
order_ct_scan	CT: non-contrast / contrast, head / spine	139
order_advanced_imaging	MRA / MRV / CTA / PET / DAT-SPECT	67
order_echocardiogram	TTE / TEE: chambers, valves, EF, thrombus	57
order_cardiac_monitoring	telemetry / Holter / loop recorder summary	57

All outputs

are Pydantic models (`.model_dump()`); each tool has a Medicare-PFS reference cost logged by CostTracker.

Hospital protocols and patient memory

Hospital protocols

YAML files at
config/hospital_rules/{hospital}/, one file
per pathway (stroke, status epilepticus, bacterial
meningitis, ...).

At session start the system prompt is augmented
with the relevant pathway rules.

Real hospitals *differ* on which panel, which contrast,
when to escalate, and the benchmark must reflect
that.

Patient memory

Per-patient long-term context: prior visits, chronic
conditions, baseline cognition, advance directives.
Reasoning should improve when the agent
remembers a 6-month gradient instead of treating
every encounter as cold.

System prompt at runtime

```
base_prompt
+ hospital_rules(pathway)
+ patient_memory(patient_id)
```

Serving stack and hardware

Models

Four open-weights backbones via **vLLM** [Kwon et al. \(2023\)](#) on **A100-40 GB**:

- Qwen3.5 with qwen3 reasoning parser plus qwen3_coder tool parser
- MedGemma with hermes tool parser
- Two further variants for ablation

Mac dev: Ollama fallback. Defaults: T=1.0, top_p=0.95, pp=1.5, max=8192.

Client and orchestration

llm/client.py strips <think> tags from Qwen and parses OpenAI-style tool calls. FastAPI port 8888 serves REST + SSE streaming. Loading via POST /api/v1/models/{key}/load (SSE progress).

Trace replay

Every run is auto-saved to data/traces/ so we can re-score without GPU time; crucial when iterating on metrics.

Training: golden trajectories → SFT

Building the SFT corpus

1. Take the case's ground-truth optimal action sequence
2. Choreograph a teacher rollout: tool call → cached output → short rationale → next call
3. Verify the trace terminates in the correct diagnosis and respects critical / contraindicated actions
4. Filter by clinician-validated cases only (κ -filtered)
5. Emit (system, user, assistant) chat tuples

Targets

Smaller deployable backbones (7–14 B). Goal: match larger frontier baselines on tool selection and diagnostic accuracy at a fraction of the inference cost.

What SFT does *not* cover

Teaches imitation of an optimal trajectory, not recovery from divergence; that's why we want RL on top.

Fine-tuning method: QLoRA on Qwen3.5-9B

What actually trains

- Base Qwen3.5-9B frozen in 4-bit NF4 (BitsAndBytes, double quant, bf16 compute)
- LoRA adapters $r=64$, $\alpha=128$ on *all* attention + MLP projections
- Only adapters update ($\sim$ tens of MB), merged for deployment; one recipe for SFT and RL

Why QLoRA: a single A100-40 GB

- Full fine-tuning needs $\sim 4\times$ the weights for optimiser states — multi-GPU only
- 4-bit base shrinks 18 GB $\rightarrow$ 5 GB, leaving room for 6144-token ReAct traces
- NF4 + double quant near-lossless: matches 16-bit LoRA quality [Dettmers et al. \(2023\)](#)

Alternatives considered

Full fine-tuning (multi-GPU only), **plain 16-bit LoRA** (no memory headroom), **DoRA** / **LoftQ** (quality upgrade, future ablation), and **prefix / prompt tuning** (too weak for tool use).

What's next: GRPO and graph-structured reasoning

GRPO: relative reasoning RL

Sample k trajectories per case, score each, optimise with rewards relative to the group mean, with no value network.

Why it fits:

- Reasoning quality is a **relative** signal across rollouts on the same case
- Our metrics (Dx acc, tool match, cost, red-herring) are already group-comparable

Both are research directions, not yet implemented. The benchmark is designed so progress on either is directly visible in the metrics.

Graph-based structured reasoning

Replace free-text Thought with a typed *clinical reasoning graph*:

nodes $\in$ {symptom, exam finding, hypothesis, evidence, action, conclusion}; edges $\in$ {supports, contradicts, triggers, depends-on}.

Why:

- Constrains hallucination at generation
- Makes reasoning **auditable** by clinicians
- Maps onto our optimal action sequence



Results



How we evaluate: six complementary metrics

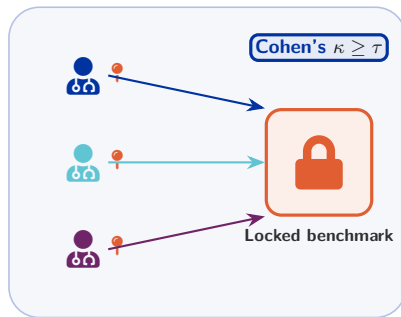
Metric	What it captures
Diagnostic accuracy	ICD-10 match against ground truth (exact + group-level)
Tool selection	precision / recall of called tools vs. optimal action sequence
Tool parameterisation	right panel / sequence / region, scored per step
Cost efficiency	sum of Medicare-PFS rates over all tool calls
Red-herring resistance	did the agent commit to a contraindicated / red-herring action?
Clinician score	aggregated verdict from the validation platform

Every metric runs against the same evaluator (`evaluation/runner.py`) with `format_patient_info()` as the single source of truth, so numbers across variants are directly comparable.

Clinician validation as the gold standard

From reviewer pool to score

1. Each case reviewed by ≥ 3 clinicians
2. Per-field annotations (note / issue / error) and a verdict (approved / needs_changes) stored on the platform
3. Admin computes Cohen's κ per field type
4. Cases with κ above threshold + consistent verdicts enter the *locked* benchmark
5. Agent runs are scored **only** on locked cases



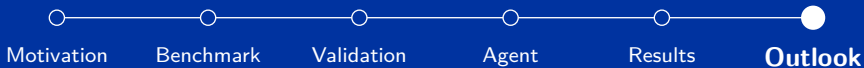
Current results: model comparison on NeuroBench

Variant	Dx %	Tool F1	Param %	Cost \$	Red-h %
Vanilla LLM (Qwen3.5)	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>
+ Tools (ReAct)	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>
+ Tools + SFT (golden)	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>
MedGemma + Tools	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>
Frontier reference	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>
+ GRPO (planned)	–	–	–	–	–
+ Graph reasoning (planned)	–	–	–	–	–

Numbers from the latest full evaluation sweep. Dx = ICD-group match; Tool F1 vs. optimal sequence; Param = correct parameters; Cost in Medicare-PFS \$/case.



Outlook



Recap and roadmap

What we've shown

- Medical LLM eval needs to move past Q&A
- NeuroBench: 500+ cases, 20 conditions, 12 tools, evidence revealed via tool calls
- Custom platform for triple-blind clinician validation with inter-rater agreement
- NeuroAgent: ReAct loop, hospital protocols, memory, cost tracking, vLLM on A100s
- Variant ladder all measured on the same metrics

Next steps (paper milestones)

- Finish clinician validation on all 516 cases
- Lock benchmark + freeze metrics
- Full sweep across the variant ladder
- Implement GRPO; prototype graph reasoning
- Draft for **Nature Machine Intelligence**

Open questions

Metric weighting? Reviewer pool size? Conditions to expand?

References

- S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao. *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR, 2023. arXiv:[2210.03629](#).
- T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, T. Scialom. *Toolformer: Language Models Can Teach Themselves to Use Tools*. NeurIPS, 2023. arXiv:[2302.04761](#).
- D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, P. Szolovits. *What Disease Does This Patient Have? A Large-Scale Open-Domain QA Dataset from Medical Exams (MedQA)*. Applied Sciences **11**(14), 6421, 2021.
- A. Pal, L. K. Umapathi, M. Sankarasubbu. *MedMCQA: A Large-Scale Multi-Subject Multi-Choice Dataset for Medical Domain QA*. CHIL, 2022. arXiv:[2203.14371](#).
- K. Singhal et al. *Towards Expert-Level Medical Question Answering with Large Language Models*. arXiv:[2305.09617](#), 2023.
- K. Wu, E. Wu, R. Thapa, K. Wei, A. Zhang, A. Suresh, J. J. Tao, M. W. Sun, A. Lozano, J. Zou. *MedCaseReasoning: Evaluating and Learning Diagnostic Reasoning from Clinical Case Reports*. arXiv:[2505.11733](#), 2025.
- S. Schmidgall, R. Ziaei, C. Harris, E. Reis, J. Jopling, M. Moor. *AgentClinic: A Multimodal Agent Benchmark to Evaluate AI in Simulated Clinical Environments*. npj Digital Medicine, 2026. arXiv:[2405.07960](#).

References (cont.)

- Z. Zhao, Q. Jin, F. Chen, T. Peng, S. Yu. *PMC-Patients: A Large-Scale Dataset of Patient Summaries and Relations*. Sci. Data **10**, 909, 2023. arXiv:[2202.13876](#).
- W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica. *Efficient Memory Management for Large Language Model Serving with PagedAttention (vLLM)*. SOSP, 2023. arXiv:[2309.06180](#).
- Z. Shao, P. Wang, Q. Zhu et al. *DeepSeekMath: Pushing the Limits of Mathematical Reasoning (introduces GRPO)*. arXiv:[2402.03300](#), 2024.
- T. Detrmers, A. Pagnoni, A. Holtzman, L. Zettlemoyer. *QLoRA: Efficient Finetuning of Quantized LLMs*. NeurIPS, 2023. arXiv:[2305.14314](#).
- A. Protani et al. *Federated GNNs for EEG-Based Stroke Assessment*. UniReps Workshop, 2024. arXiv:[2411.02286](#).



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